

Program 4: Integrate git with docker for source-controlled application builds

File structure:

/Dockerfile

/package.json

/package-lock.json

/index.js

Create folder and initialize package.json

mkdir program4 > cd program4 > npm init -y > npm i express

Step 1:

Create a folder for prg4

Mkdir program4

Step2:

Created a [index.js](#) file

```
const express = require('express')
const app = express()
app.get('/',(req,res)=>res.send("Hello from 3rd program"))
app.listen(3000,()=>{console.log("Server running on http://localhost:3000");})
```

Step 3:

Created a Dockerfile

```
#stage 1
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
#stage 2
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node","index.js"]
```

Step 4: build and run the image

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program4$ docker build -t program4 .
[+] Building 4.6s (12/12) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 221B
=> [internal] load metadata for docker.io/library/node:18-slim
=> [internal] load metadata for docker.io/library/node:18
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [build 1/3] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
=> [stage-1 1/3] FROM docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651838830e
=> [internal] load build context
=> => transferring context: 2.75MB
=> CACHED [build 2/3] WORKDIR /app
=> [build 3/3] COPY . .
=> CACHED [stage-1 2/3] WORKDIR /app
=> [stage-1 3/3] COPY --from=build /app /app
=> exporting to image
=> => exporting layers
=> => writing image sha256:f041719215de337e37807f3a9b75e1496d3ebf29307bb6c41edcd6be535379a8
=> => naming to docker.io/library/program4
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program4$ █
```

Step 5:

Go to github and create a repo and get a PAT

Inside the folder

```
git init -y
git add .
git commit -m "initial commit"
git branch -M main
```

```
git remote add origin https://github.com/<username>/<repo>.git
```

git push -u origin main

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program4$ git add .
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program4$ git commit -m "Intial commit"
[main e3c1a1e] Intial commit
633 files changed, 1 insertion(+), 71934 deletions(-)
create mode 100644 .gitignore
delete mode 100644 node_modules/.package-lock.json
delete mode 100644 node_modules/accepts/HISTORY.md
delete mode 100644 node_modules/accepts/LICENSE
delete mode 100644 node_modules/accepts/README.md
delete mode 100644 node_modules/accepts/index.js
delete mode 100644 node_modules/accepts/package.json
delete mode 100644 node_modules/body-parser/HISTORY.md
delete mode 100644 node_modules/body-parser/LICENSE
delete mode 100644 node_modules/body-parser/README.md
delete mode 100644 node_modules/body-parser/index.js
delete mode 100644 node_modules/body-parser/lib/read.js
delete mode 100644 node_modules/body-parser/lib/types/json.js
delete mode 100644 node_modules/body-parser/lib/types/raw.js
delete mode 100644 node_modules/body-parser/lib/types/text.js
delete mode 100644 node_modules/body-parser/lib/types/urlencoded.js
delete mode 100644 node_modules/body-parser/lib/utils.js
delete mode 100644 node_modules/body-parser/package.json
delete mode 100644 node_modules/bytes/History.md
delete mode 100644 node_modules/bytes/LICENSE
delete mode 100644 node_modules/bytes/Readme.md
```

Step 6: know we can see the files inside the repo

The screenshot shows the GitHub repository page for 'program4' by user 'ajay-m-l'. The repository is public and has 2 commits. The commit history table shows the following files:

File	Commit	Time
.gitignore	Intial commit	19 minutes ago
Dockerfile	Intial commit	31 minutes ago
index.js	Intial commit	31 minutes ago
package-lock.json	Intial commit	31 minutes ago
package.json	Intial commit	31 minutes ago

Step 7: verify in the localhost//3000

The screenshot shows a web browser window with the address bar displaying 'http://localhost:3000'. The page title is 'Docker+Git integration'. The browser tabs include 'Firewall Authentication Keep', 'ajay-m-l/program4', '[GitHub] Sudo email verifi', and 'localhost:3000/'.